

where: r_c = compression ratio
 $R = r/a$

The cross-sectional area of a cylinder and the surface area of a flat-topped piston are each given by:

$$A_p = (\pi/4)B^2 \quad (2-15)$$

The combustion chamber surface area is:

$$A = A_{ch} + A_p + \pi B(r + a - s) \quad (2-16)$$

where A_{ch} is the cylinder head surface area, which will be somewhat larger than A_p .

Then if the definitions for r , a , s , and R are used, Eq. (2-16) can be rewritten as:

$$A = A_{ch} + A_p + (\pi BS/2)[R + 1 - \cos\theta - \sqrt{R^2 - \sin^2\theta}] \quad (2-17)$$

EXAMPLE PROBLEM 2-1

John's automobile has a three-liter SI V6 engine that operates on a four-stroke cycle at 3600 RPM. The compression ratio is 9.5, the length of connecting rods is 16.6 cm, and the engine is square ($B = S$). At this speed, combustion ends at 20° aTDC. Calculate:

1. cylinder bore and stroke length
2. average piston speed
3. clearance volume of one cylinder
4. piston speed at the end of combustion
5. distance the piston has traveled from TDC at the end of combustion
6. volume in the combustion chamber at the end of combustion

1) For one cylinder, using Eq. (2-8) with $S = B$:

$$V_d = V_{total}/6 = 3L/6 = 0.5 L = 0.0005 m^3 = (\pi/4)B^2S = (\pi/4)B^3$$

$$\underline{B = 0.0860 m = 8.60 cm = S}$$

2) Using Eq. (2-2) to find average piston speed:

$$\bar{U}_p = 2SN = (2 \text{ strokes/rev})(0.0860 m/stroke)(3600/60 \text{ rev/sec})$$

$$\underline{= 10.32 m/sec}$$

3) Using Eq. (2-12) to find the clearance volume of one cylinder:

$$r_c = 9.5 = (V_d + V_c)/V_c = (0.0005 + V_c)/V_c$$

$$\underline{V_c = 0.000059 m^3 = 59 cm^3}$$

4) Crank offset, $a = S/2 = 0.0430 m = 4.30 cm$

$$R = r/a = 16.6 cm/4.30 cm = 3.86$$

Using Eq. (2-5) to find instantaneous piston speed:

$$\begin{aligned} U_p/\bar{U}_p &= (\pi/2) \sin \theta [1 + (\cos \theta / \sqrt{R^2 - \sin^2 \theta})] \\ &= (\pi/2) \sin (20^\circ) \{1 + [\cos (20^\circ) / \sqrt{(3.86)^2 - \sin^2 (20^\circ)}]\} \\ &= 0.668 \end{aligned}$$

$$U_p = 0.668 \bar{U}_p = (0.668)(10.32 \text{ m/sec}) = \underline{6.89 \text{ m/sec}}$$

5) Using Eq. (2-3) to find piston position:

$$\begin{aligned} s &= a \cos \theta + \sqrt{r^2 - a^2 \sin^2 \theta} \\ &= (0.0430 \text{ m}) \cos (20^\circ) + \sqrt{(0.166 \text{ m})^2 - (0.0430 \text{ m})^2 \sin^2 (20^\circ)} \\ &= 0.206 \text{ m} \end{aligned}$$

Distance from TDC:

$$\begin{aligned} x &= r + a - s = (0.166 \text{ m}) + (0.043 \text{ m}) - (0.206 \text{ m}) \\ &= \underline{0.003 \text{ m} = 0.3 \text{ cm}} \end{aligned}$$

6) Using Eq. (2-14) to find instantaneous volume:

$$\begin{aligned} V/V_c &= 1 + \frac{1}{2}(r_c - 1)[R + 1 - \cos \theta - \sqrt{R^2 - \sin^2 \theta}] \\ &= 1 + \frac{1}{2}(9.5 - 1)[3.86 + 1 - \cos (20^\circ) - \sqrt{(3.86)^2 - \sin^2 (20^\circ)}] \\ &= 1.32 \end{aligned}$$

$$V = 1.32 V_c = (1.32)(59 \text{ cm}^3) = \underline{77.9 \text{ cm}^3 = 0.0000779 \text{ m}^3}$$

This indicates that, during combustion, the volume in the combustion chamber has only increased by a very small amount and shows that combustion in an SI engine occurs at almost constant volume at TDC.

2-2 WORK

Work is the output of any heat engine, and in a reciprocating IC engine this work is generated by the gases in the combustion chamber of the cylinder. Work is the result of a force acting through a distance. Force due to gas pressure on the moving piston generates the work in an IC engine cycle.

$$W = \int F dx = \int P A_p dx \quad (2-18)$$

where: P = pressure in combustion chamber

A_p = area against which the pressure acts (i.e., the piston face)

x = distance the piston moves

and: